

## Model Performance Analysis

After training all classification models on the SMOTE-balanced dataset and evaluating them on the untouched validation set, the Random Forest Classifier was the best-performing model. It achieved the highest scores across all pages, including Accuracy (0.999772), Precision (1.000000), Recall (0.999544), F1-score (0.999772), and ROC-AUC (1.000000). Random Forest performed super good, because it handles nonlinear relationships, interacts effectively with engineered features such as temperature range and seasonal indicators, and is mute to noise in meteorological data.

The Voting Ensemble, built using the top three models, also performed well, having a perfect ROC-AUC of 1.000000 and very high accuracy. This confirms that combining strong base learners improves the foundation.

The worst-performing model was the K-Nearest Neighbors (KNN) classifier. KNN had perfect recall, but its overall accuracy (0.966446) and precision (0.937126) were much lower than the other models. KNN struggles with high-dimensional data, is sensitive to scaling, and is not well-suited for meteorological variables.

The Bayesian Ensemble (GaussianNB) also underperformed due to its assumption of feature independence, which does not hold in weather datasets.

Overall, the Random Forest model provides the most reliable and accurate predictions for heatwave classification in this dataset.